ExoScout Critical Design Review

Team Name: ExoScout

www.cansat.eu

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1 Changelog

CDR

- We have decided to focus on the soil intake in our secondary mission and try to add other sensors only when the mechanism is ready

2 Introduction

2.1 Team organisation and roles

Team consists of 6 students from V High School in Bielsko-Biała, where we pursue extended programmes in math, physics and computer science. Although all of us realise the same extended subjects we present a wide variety of interdisciplinary interests and abilities. We see the CanSat competition as a great opportunity to broaden our knowledge and get experience in this type of team work. On top of that we are supported by our teacher and one of our older schoolmates, who is our mentor.

[redacted] - teacher:

- Background: [redacted]
- Field of work: Supervision of the team's progress.

[redacted] - mentor:

- Background: [redacted]
- Field of work: Consultation.
- Hours dedicated weekly: 1 hour for consultations

Karol Synowiec - team leader:

- Background: Interested in math, physics and engineering. He likes understanding how machines work and is eager to explore new science achievements especially those concerning space exploration and physics related to it. He participates in contests such as Math Olympiad, Physics Olympiad, Diament AGH and more.
- Fields of work: CanSat 3D modelling in Fusion 360, electronics and team leader

[1] https://spacecamp.no/

- Hours dedicated weekly: 10 hours

[redacted] - team member:

- Background: [redacted]
- Field of work: Software concerning sensors
- Hours dedicated weekly: 12 hours

[redacted] - team member:

- Background: [redacted]
- Field of work: Mechanical system development and design
- Hours dedicated weekly: 5 hours

[redacted] - team member:

- Background: [redacted].
- Fields of work: Recovery system, CanSat assemblement.
- Hours dedicated weekly: 10 hours

[redacted] - team member:

- Background: [redacted]
- Fields of work: software concerning soil intake, electronics.
- Hours dedicated weekly: 5 hours

[redacted] - team member:

- Background: [redacted]
- Fields of work: outreach, promotion
- Hours dedicated weekly: 5 hours

2.2 Mission objectives

2.2.1 Our vision

In our project we want to create CanSat which will act as a “scout” for a mission looking for a planet that can be settled by humans. After first assessments of a planet's potential habitability from space, a CanSat like device will be dropped on the planet's surface and will carry out further measurements of factors crucial to the life of the humankind. Data collected from primary and secondary missions will be used with a view to analyse multiple factors, which are considered to be vital for planetary habitability.

2.2.2 Primary mission

Primary mission is to measure air temperature and air pressure and to transmit it as a telemetry once per second to the ground station. Data collected from it will be used to assess if water can be liquid on the planet’s surface[2]. Temperature limit for life[3] is in the range of 0 to 113 degrees Celsius, which will be also evaluated based on those measurements.

2.2.3 Secondary mission

Secondary mission will collect more specific data about the planet. In order to do that we divide it into the two main stages:

1. During fall CanSat will measure factors concerning planetary habitability:
   - Gravitational acceleration - Too high value of it makes any motion harder, blood can’t freely flow and bones can be broken due to overload[4][5]. On the other hand low gravity implies low escape velocity, it results in molecules leaving the planet more easily and destroying its atmosphere[6].

[2] Liquid water importance from abstract of: https://pubmed.ncbi.nlm.nih.gov/11539468/
[3] Table 1 on page 185 http://www.geology.wisc.edu/astrobiology/docs/Lammer_et_al_2009_Astron_Astro_Rev.pdf
[4] https://www.astronomy.com/science/whats-the-maximum-gravity-we-could-survive/
[5] https://arxiv.org/pdf/1808.07417.pdf,
[6] extract about mass and size and hence planets gravity from https://en.m.wikipedia.org/wiki/Planetary_habitability

- Cosmic radiation (UV) - Radiation is both helpful and harmful[7]. UV radiation takes part in biosphere processes but also affects human health. Too much of it can result in cancer but it is crucial for vitamin D3 synthesis in the body.
- Magnetic field - without proper magnetic field the planet is not protected from stellar wind and cosmic radiation, which destroys its atmosphere and puts life at risk of being bombarded with ionized particles[8].
- Atmosphere O2 content - oxygen is vital to life as we know it. The reduction of oxygen provides one of the largest free energy release per electron transfer[9], so it is a great source of energy for complex life.

2. Vertical landing and taking a soil sample with Archimedes screw-like device[10] to measure its humidity:
    - Analysis of soil helps in the search for past water activities. Mineralogy of it is important in determining the processes that created the planet. Soil is widely analysed by rovers (f.e. on Mars)[11].

2.2.4 Assessment

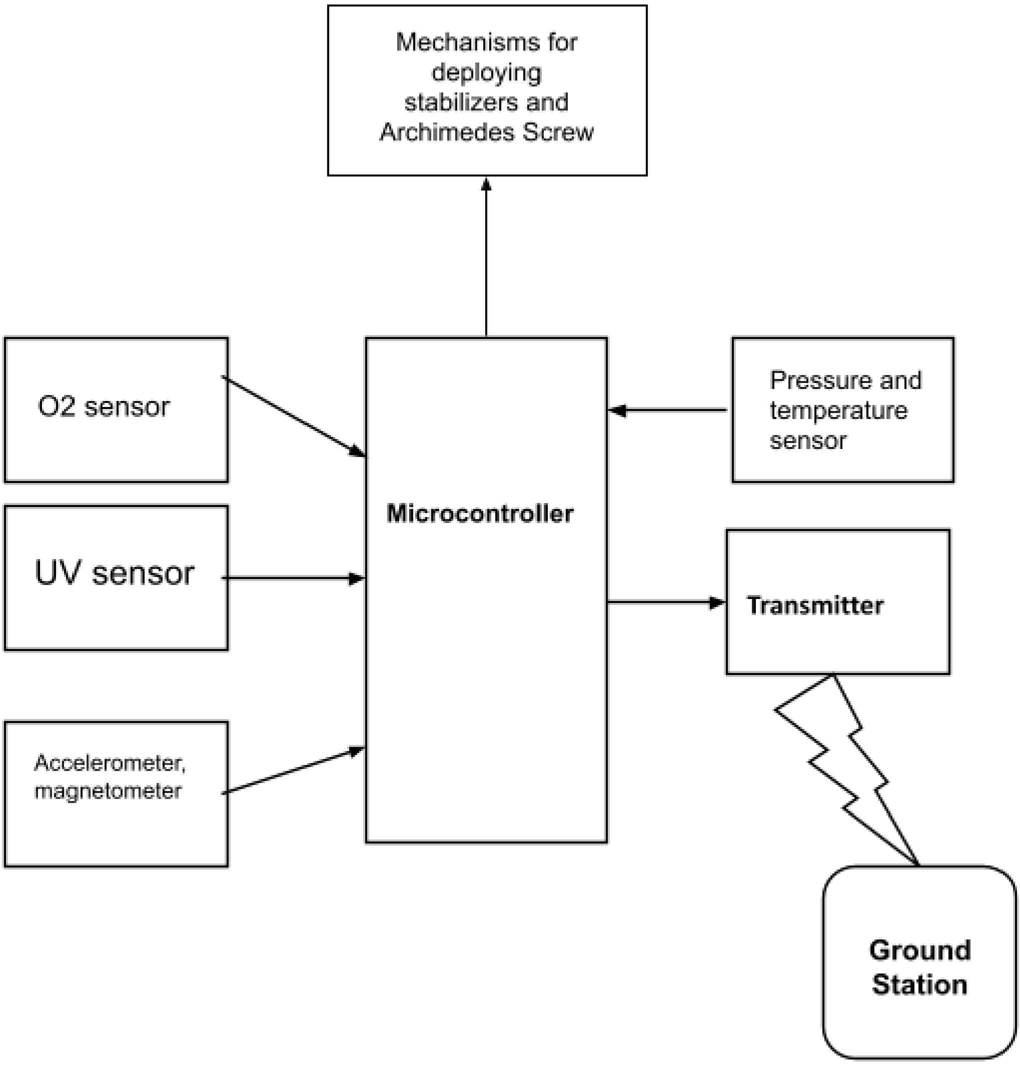
In conditions of the launch campaign on Earth we will test the measurements correctness by comparing them with those measured by more sophisticated devices. From the mechanical side we want to test if we can make CanSat land vertically by reducing its speed and using side stabilizers.

[7] https://www.cdc.gov/nceh/features/uv-radiation-safety/index.html
[8] Significance part from: https://en.wikipedia.org/wiki/Earth%27s_magnetic_field
[9] https://earthscience.rice.edu/wp-content/uploads/2015/08/Catling-Astrobiology-2005.pdf
[10] https://en.wikipedia.org/wiki/Archimedes%27_screw
[11] https://mars.nasa.gov/resources/4910/inspecting-soils-across-mars/

3 CANSAT DESCRIPTION

3.1 Mission overview

After launch from a rocket at a destined altitude (1500m-2500m) CanSat will deploy a parachute and side stabilizers. During the fall the device will start its measurements and reach terminal velocity of around 6 m/s in order to land vertically which will enable us to collect soil with Archimedes Screw-like mechanism after landing.



3.2 Mechanical/structural design

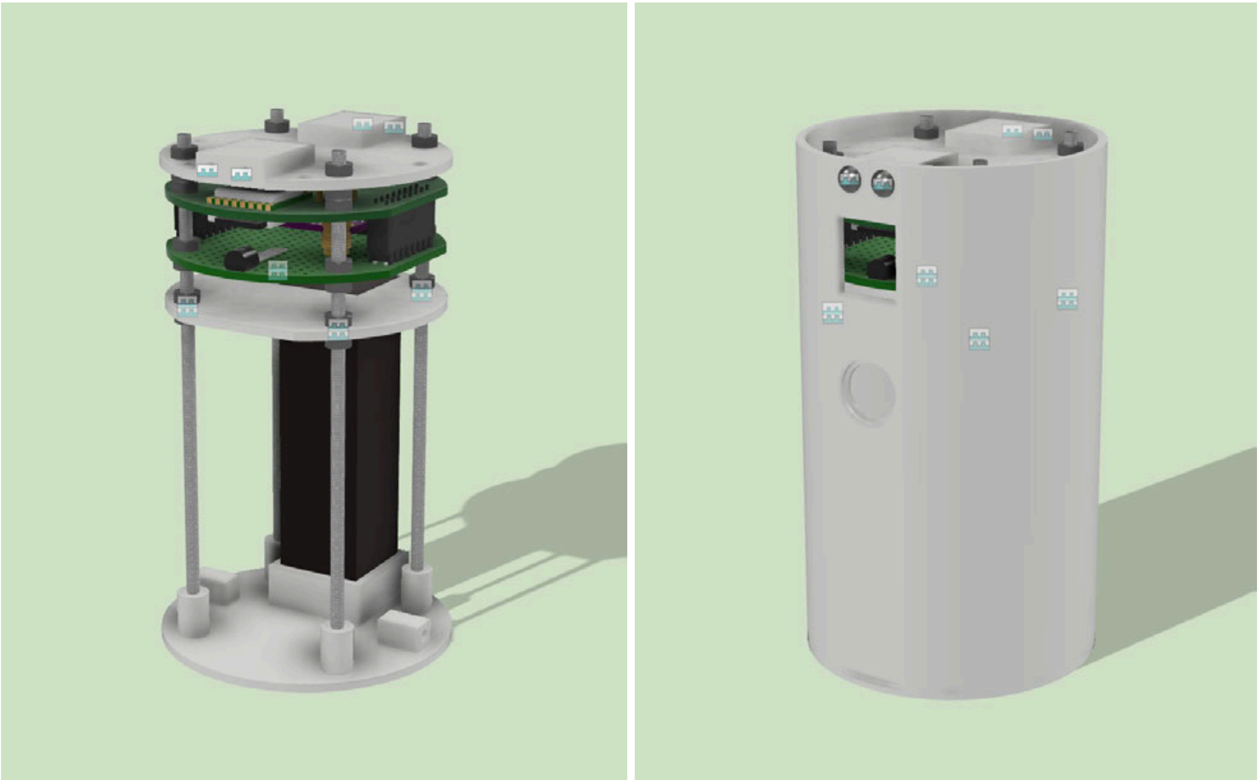
CanSat and soil intake mechanism have been redesigned since PDR.
Upper part is a place for a computer, prototype board and sensors:

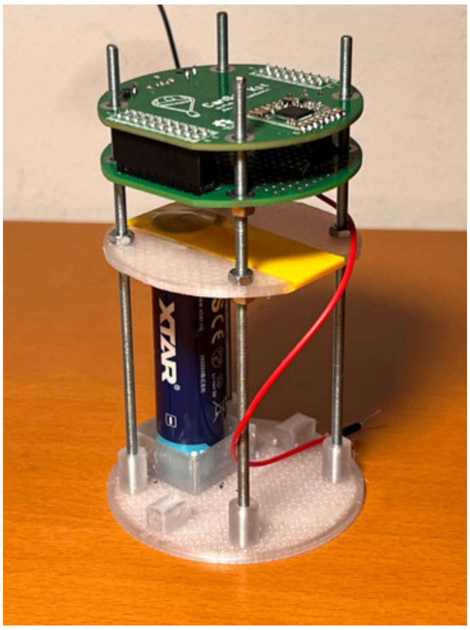
- Cap has holes for mounting the parachute and leading out the antenna, it also has mounting places for the cover
- Sensors are soldered on the prototype board

Lower part is a place for the battery and soil intake mechanism.

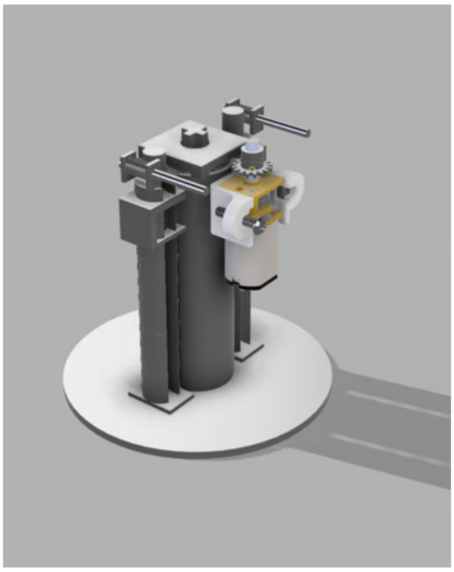
- Shelf's function is to stabilize and keep the battery in place
- Base has a mount for the battery

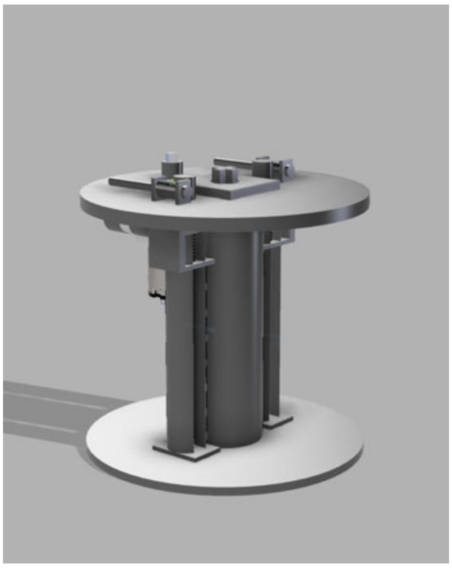
CanSat's cover and some internal parts were printed on a 3D printer with PETG filament. Cap, Arduino, prototype board and shelf are rigidly mounted on M3 stainless steel rods with M3 nuts. The mount for the battery will be redesigned to take less space and fit the cell holder. The cover has holes to access the programming socket and microSD card. There is also a place for the switch, which will be delivered soon.





Soil intake mechanism has been redesigned, but we still need to try to simplify and miniaturize it since there is less space in CanSat than we thought. Soil collecting mechanism is powered by one Pololu HPCB 10:1 motor. It propels the screw directly by a cross-shaped shaft, which goes all the way down to the very bottom, so that the force is applied through the whole range of movement of the lift. Speaking of the lift, iti is propelled by two worm gears enclosed in columns with proper threading on the inside, thus converting rotary motion to sliding movement. Worm gears drive works on the same principle as the screw itself, accommodating Allen-Key like shaft device to propel them.



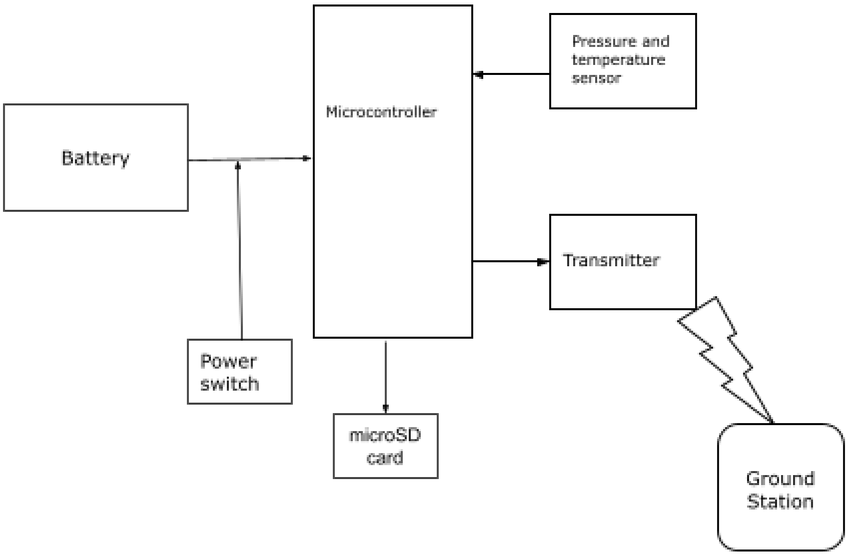


The measured mass so far (including parachute) is 198 g, it will be bigger after adding secondary mission elements.

3.3 Electrical design

3.3.1 General architecture and main devices

Figure showing the general design of electrical system so far:



- CanSat kit Arduino board

3.3.2 Primary mission devices

- BMP 280 sensor
- LM35 sensor
- 16 GB microSD card

3.3.3 Secondary mission devices

- Pololu 3071 HPCB 10:1 motor
- Motor driver L293D

3.3.4 Power supply

Estimated power consumption:

- Microcontroller and radio - 700 mW

- Sensors - <20 mW
- Motor - 16000 mW

Assuming that the motor will work for the maximum of 10 minutes and other devices should work for 6 hours we get the final capacity of around 7166 mWh. The Li-ion battery has a voltage of 3.7 V so to have a safe margin 2600 mAh power cell is suitable.

- Li-Ion XTAR 2600 mAh power cell
- On-Off power switch, which will be delivered soon: https://botland.store/rotary-switches/6357-switch-spst-on-off-ignition-switch-with-keys-5904422368180.html

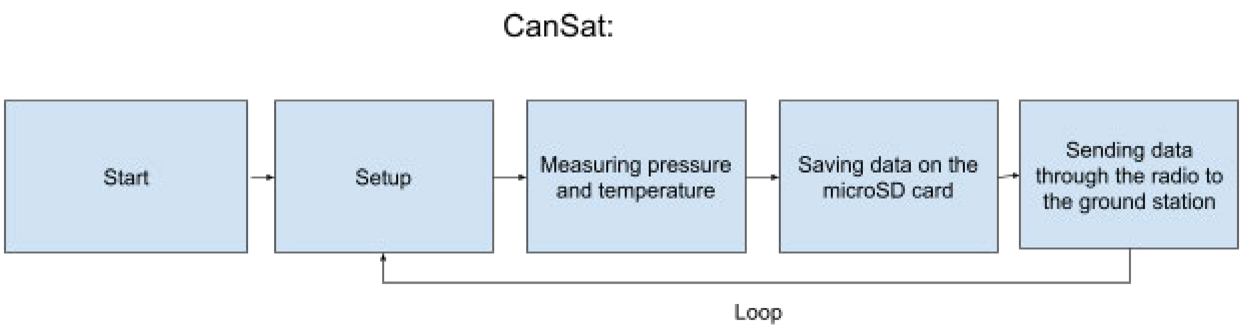
3.3.5 Communication system

Communication system will only send data from CanSat to the ground station.

- SX1278 module from CanSat kit board
- Flexible 14 cm antenna

3.4 **Software design**

In primary mission software should plot measured temperature and pressure as a point on phase diagram[12] in order to check if water is liquid in these conditions.
In the secondary mission software will check if measured factors are in range optimal for life and send information to ground stations. Data should be live displayed on laptops with use of software like Grafana[13]. Program will also launch side stabilisers after drop from the racket and start the Archimedes screw after landing.
The code so far is available there: https://github.com/LoloxonCc/ExoScout
Schematic description of how it works:



To develop our project we use Arduino IDE. We are programming in C++. During the test of the battery CanSat gathered around 1 MB of data during 7 hours of tests.

[12] https://openstax.org/books/chemistry-2e/pages/10-4-phase-diagrams
[13] https://grafana.com/

3.5 Recovery system[14]

Parachute will be made out of nylon and will have a hemispherical shape. It will be attached to the CanSat with four lines. Its area can be obtained from the equation for equilibrium of forces:

$$mg = \frac{1}{2} C_D \rho A v^2$$

Where: m - mass of the CanSat, g - Earth acceleration, C_D - drag coefficient of parachute, ρ - density of air and v is terminal velocity.

After some careful conversions we get:

$$A = \frac{2mg}{C_D \rho v^2}$$

Substituting values:

$g=9.81\ m/s^2$, $C_D=0.62$, $\rho=1.204\ kg/m^3$, $v=6\ m/s$, m in range of 0.3 to 0.35 kg

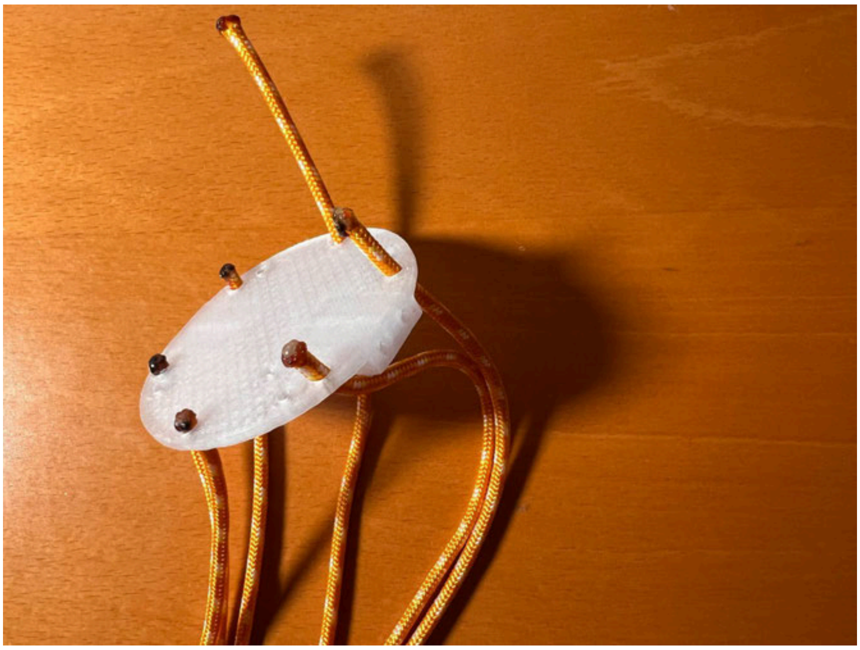
We get:

$$0.219\, m^2 \le A \le 0.255\, m^2$$

Shape of hemispherical parachute and our parachute:

Parachute is mounted to the cap of CanSat like that:

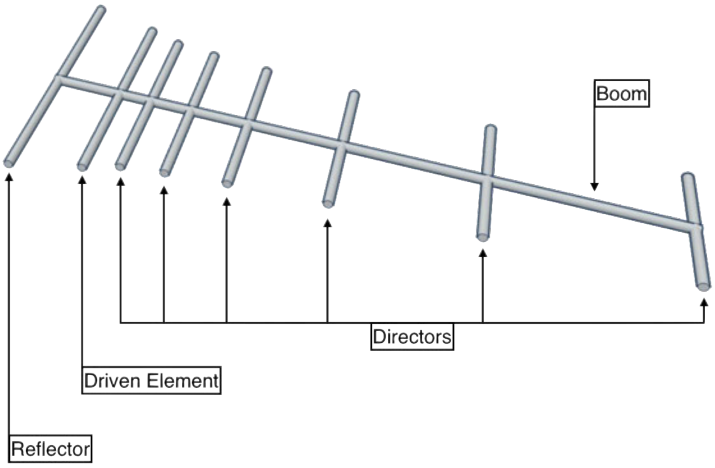
[14] https://cansat.esa.int/design-your-parachute/



3.6 **Ground support equipment**[15]

Data from the CanSat will be transmitted to the ground station. Equipment concerned with these will consist of:

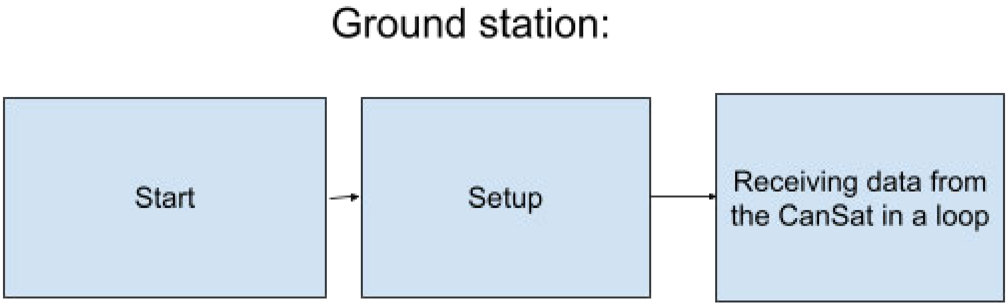
- Yagi-Uda antenna:



[15] https://cansat.esa.int/communicating-with-radio/

- Computer to display received data
- Arduino board from CanSat kit for communication of antenna and the computer

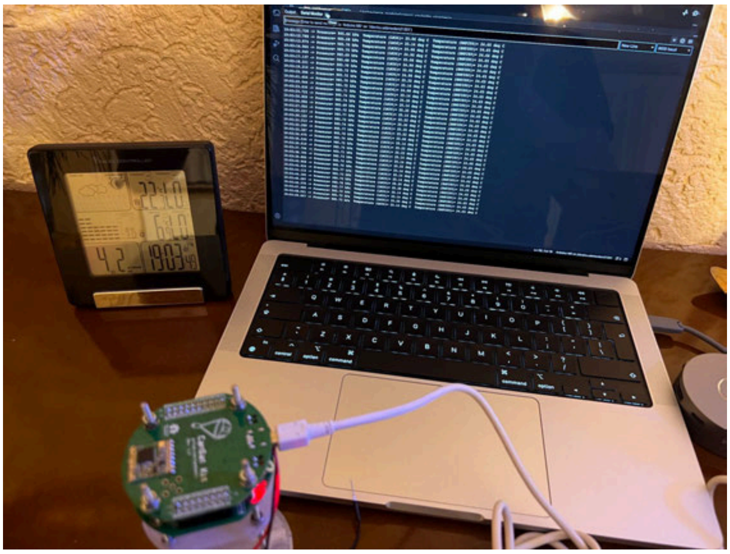
Schematic of software working:



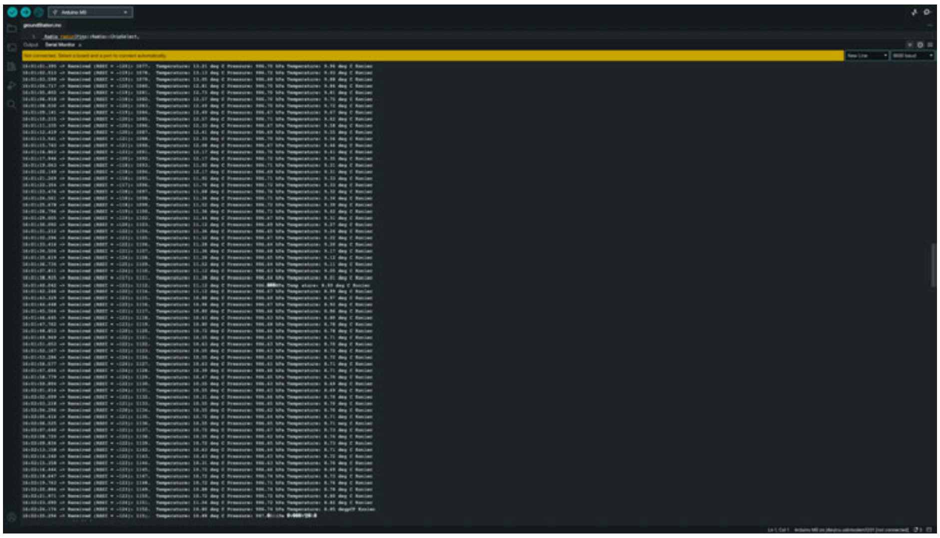
4 Test Campaign

4.1 Primary mission tests

Temperature and pressure measurements have been conducted indoor and outdoor. During indoor tests we compared the results with measurements of other measuring device. The difference between the measurements can be a result of the difference between sensors.



Outdoor test has been carried out together with a test of radio communication and compared with the weather forecast. The difference between the forecast and the measured parameters can be a result of more general character of forecast and more local character of the measurement.



All of the measurements have been saved on a microSD card.

4.2 Secondary mission tests

Secondary mission tests are going to be conducted in the coming time. The intake mechanism will be tested on different grounds and with different systems of stabilisation.

4.3 Tests of recovery system

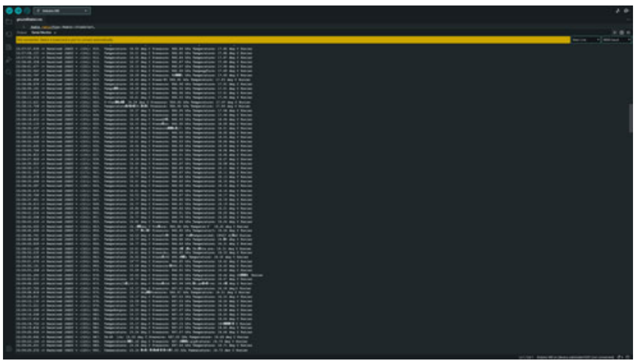
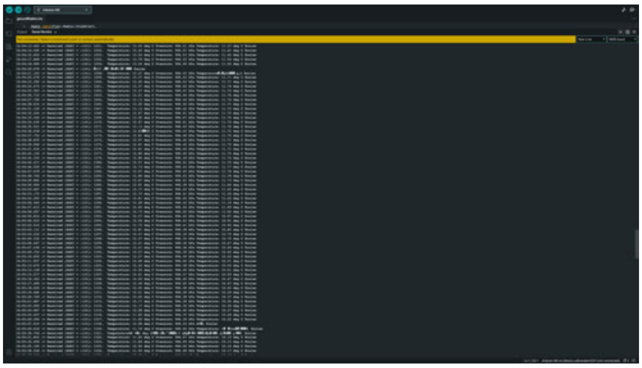
Recovery system has been tested on a dummy mass model that was loaded to 350 grams. The parachute opened properly and let the can perform a safe landing. Link to the film:
https://drive.google.com/file/d/1jJeGuMxGkSwwdG5PPjA7YzrF0h9I-vKA/view?usp=sharing
The test flight was conducted on a low height, so further testing needs to be done on higher buildings. We also need to estimate the velocity.

4.4 Communication system range tests

Communication system was tested on multiple distances. First test was conducted indoors without additional antennas to check if it is working properly. Later tests were carried out outside in the suburbs of a city. The neighbourhood is full of houses and shops so the transmission could be interrupted but it did great. The test has been conducted on four distances:

- 500 metres - signal was uninterrupted, there was no loss of transmitted data, RSSI value was around -50
- 1 kilometre - signal was uninterrupted, there was no loss of transmitted data, RSSI value was around -90
- 1.5 kilometre - signal was rarely interrupted, some of the frames were lost, but it was only a fraction of the successfully sent data, RSSI value was around -110
- 2 kilometre - signal was more often interrupted, around 1 in 20 frames were lost or damaged, RSSI value was around -120

In the environment of houses and many sources of interference we reckon that our tests were rather successful. Further testing will be conducted on more open spaces without so many disruptions and on longer distances.

4.5 **Energy budget tests**

Energy budget has been tested with CanSat:

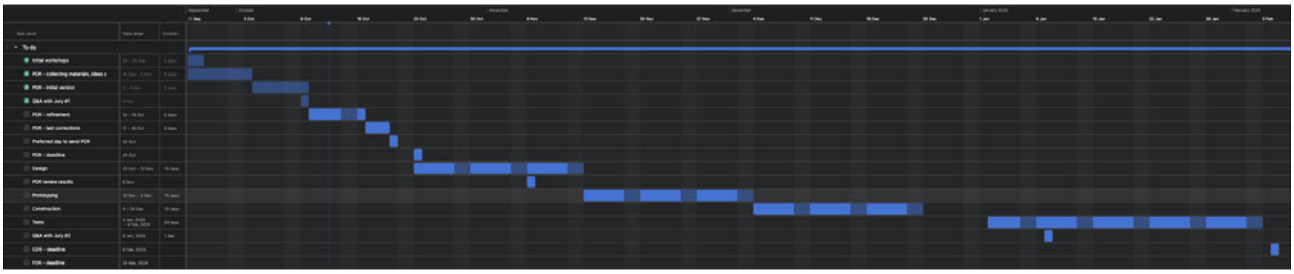
- Measuring temperature and pressure
- Saving it on microSD card
- Transmitting it by the radio communication

The last measurement has been saved after around 24000 seconds which is around 6 hours 45 minutes. Further testing will include testing the battery with motor enabled for 10 minutes and measurements for the whole test.

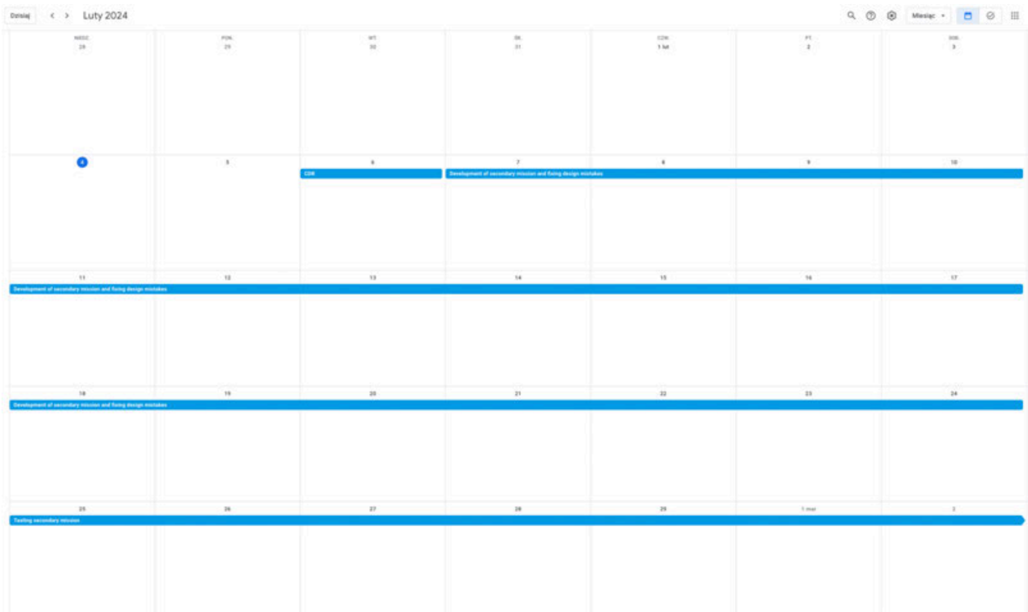
5 Project Planning

5.1 **Time schedule**

We decided to prepare PDR in 4 one week intervals, first was for collecting information and sources for the report, second was for writing the initial version of it, third for making it more detailed and the last one for last corrections. After PDR we plan to schedule work, on each phase of the project, in 3 week-long intervals. As of now we separated 5 main stages, namely: design, prototyping, construction, tests and writing CDR. They will consist of subparts. We make a rather strict schedule (tests should start before Q&A with Jury on 9 January 2024) to make sure that even if we fail to keep some dates we can take one more week to work better on some stage of the project. Materials to write CDR such as: notes, designs, etc. will be collected during all phases and consequently put together into the report. Gantt chart presenting our schedule until CDR (from Asana):



CDR has been written on the base of notes taken during work, tests and our thoughts about the project. The prediction about more time being needed for work on some stages were right and we failed to work on some parts of the project yet. To address areas that we overlooked so far we plan to work on them in the coming time before FDR. Our schedule:



5.2 Task list

Tasks done before PDR:

- Writing PDR:
  - Sources and ideas collection
  - Primary and secondary mission factors description
  - Initial version of PDR
  - Refinement of PDR
  - Last corrections to PDR
- Outreach:
  - Instagram account setup and first posts
- Design of CanSat:
  - First 3D model of CanSat
  - Initial choice of components

Tasks done before CDR:

- Design:
  - 3D model of CanSat with all details
  - Parachute design
  - Ground station design
  - Electrical design
  - Detailed description of software
- Prototyping:
  - Writing software
  - Checking if parts work alone
  - Making a parachute

- Construction:
    - Assembling all parts into a CanSat
    - Mounting a parachute
    - Making ground station
- Tests:
    - Primary mission sensors check
    - Radio communication test
    - Parachute test
    - Troubleshooting
- Outreach:
    - Posts
- Budget:
    - Looking for sponsors

Tasks to be done before FDR:

- Design:
    - fix some mistakes in model
        - battery mount
        - screws mounting
        - cover shape
        - more tolerance on dimensions
- Secondary mission:
    - work out the feasibility of the intake mechanism and design it to be possible to include in the can
    - add sensors
    - consider other factors that can be evaluated in our mission
- Tests:
    - further tests of radio communication
    - further tests of parachute
    - tests of secondary mission
    - further tests of energy budget
- Outreach:
    - make more content on social media
    - lead workshops
- Writing the FDR

5.3 Resource estimation

5.3.1 Budget

PDR:

Estimation of the costs:

- UV sensor - 6.50 EUR
- O2 sensor - 97.50 EUR

- Accelerometer, magnetometer - 19.90 EUR
- Elements from CanSat kit - 50 EUR
- motor - 24.50 EUR

These will be the most expensive parts. Even after adding other parts (filament, screws, etc.), the budget should not exceed 300 EUR.
CDR:
Costs so far:

Element:	Cost:
Mainboard (CanSat kit)	40.51 EUR
Prototype board (CanSat kit)	4.63 EUR
Temperature sensor LM35 (CanSat kit)	5.32 EUR
Pressure sensor BMP280 (CanSat kit)	2.32 EUR
Motor	28.92 EUR
Motor driver	2.98 EUR
Power cell	6.90 EUR
Cell holder	0.90 EUR
MicroSD card	2.90 EUR
Switch	0.50 EUR
Filament	18.50 EUR
Ripstop nylon	19.46 EUR
Cord	6.95 EUR
Threaded rod, bolts and screws	4 EUR
Summary:	144.79 EUR

5.3.2 External support

List of companies and organisations that provide us with sponsorship or other support:

- ASTOR
- EnCo

- University of Bielsko-Biała

6 OUTREACH PROGRAMME

Even though space exploration is a hot topic now, there are still a lot of people who do not know a lot about it. In our outreach programme we want to address them and people already fascinated about the subject, so we will create content appropriate for different groups of people. In order to achieve these goals we plan actions such as:

- regular posting on social media about progresses and project objectives (Instagram, Facebook),
- own website,
- meetings with other students group to talk about space exploration and our project,
- live transmission of CanSat descent during Launch Campaign,
- presenting project on science festivals

We plan to make posts about:

- CanSat competition
- Our primary and secondary mission
- Our team
- Progresses and achievements

So far we have designed our logo which is displayed on the first site of the document and we have set up Instagram account where we started posting about our project:
https://instagram.com/exoscoutteam?igshid=MzRlODBiNWFlZA==
We also participated in a science conference "Engineer of 21st century" on University of Bielsko-Biała.